



Homework: Rationalization and Algebraic Fractions

Name: _____

Group: IGCSE

You must write all your procedures in pencil. Your procedures must be written in a clear and orderly manner to obtain full marks. The maximum score is 44 points.

Score: _____ / 44

EXERCISE 1 (Rationalization). Solve the following fractions:

[19 pts]

Steps to Rationalize

1. Identify which case each fraction corresponds to.

3. Add the fractions by finding the least common multiple (LCM) of the denominators.

2. Rationalize each fraction.

4. Simplify the resulting fraction, if possible.

1. $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{8}}$

3. $\frac{3}{\sqrt{5}+1} + \frac{2}{\sqrt{5}-1}$

5. $\frac{2}{\sqrt{6}} + \frac{3}{\sqrt{24}} - \frac{1}{\sqrt{3}}$

2. $\frac{5}{\sqrt{3}} - \frac{2}{\sqrt{12}}$

4. $\frac{1}{\sqrt{7}-\sqrt{3}} - \frac{1}{\sqrt{7}+\sqrt{3}}$

6. $\frac{\sqrt{18}}{\sqrt{2}+\sqrt{3}} + \frac{\sqrt{8}}{\sqrt{3}-\sqrt{2}}$

EXERCISE 2 (Algebraic Fractions). Solve the following fractions:

[20 pts]

Steps to Add or Subtract Algebraic Fractions

1. Multiply the denominators to get a common denominator.
2. Cross-multiply to adjust the numerators.
3. Add the numerators and simplify the fraction if possible.

1. $\frac{2x^2y}{2x^2y^2} + \frac{z}{3xyz}$

3. $\frac{4m^2n}{m^2-n^2} + \frac{n^2}{m-n}$

5. $\frac{x^3}{x^2y-y^2z} - \frac{y^2}{xy-yz}$

2. $\frac{a^3}{2a^2b} - \frac{b^2}{3bc}$

4. $\frac{2q^2rp}{p^2qr^3} + \frac{r}{q}$

EXERCISE 3 (Previous Topics). *Simplify the following expressions:*

[5 pts]

1. $\sqrt{x^{10}} \cdot \sqrt[3]{x^6}$

3. $\sqrt{72} + \sqrt{50}$

5. $\sqrt[3]{54} + \sqrt[3]{16}$

2. $\frac{x^8 \cdot \sqrt{x^4}}{\sqrt[3]{x^6}}$

4. $\frac{2q^2r}{p^2qr^3}$